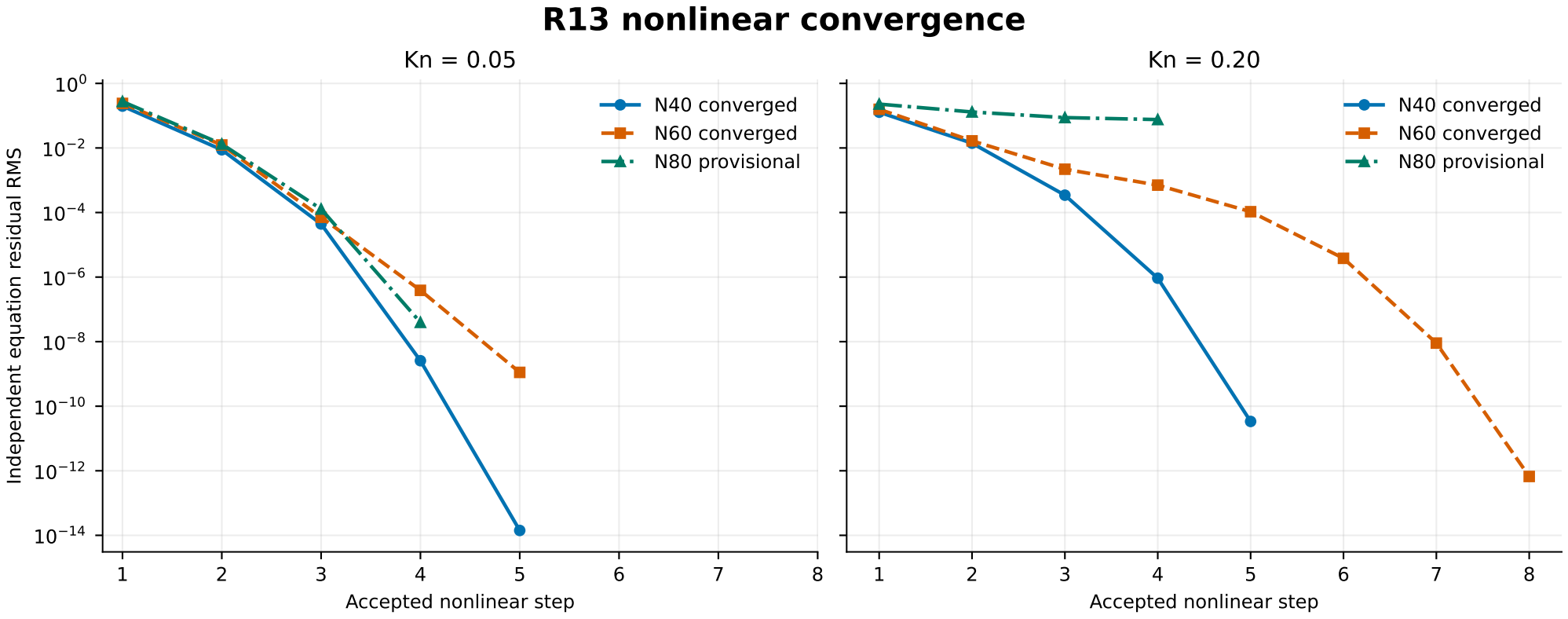
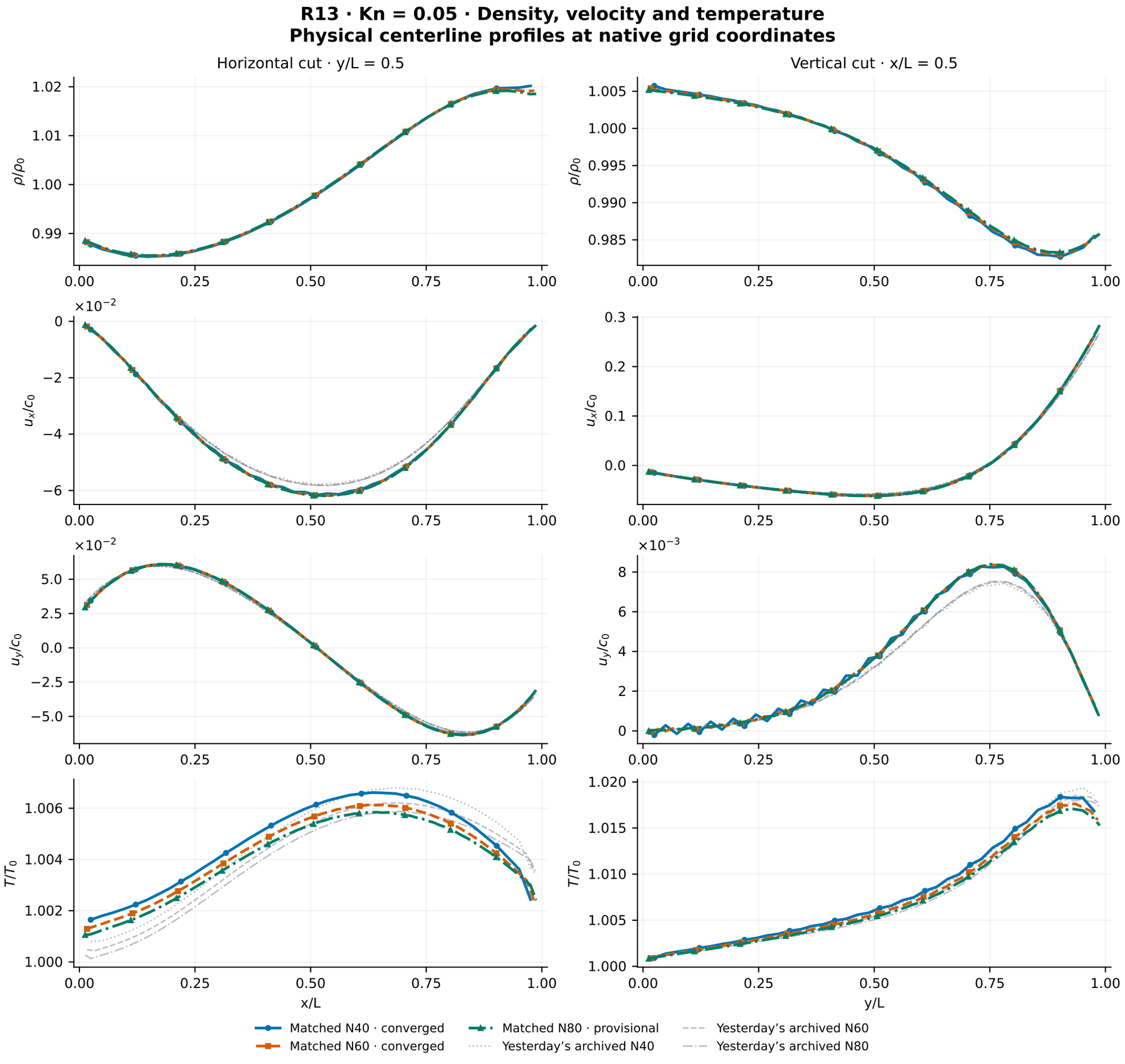
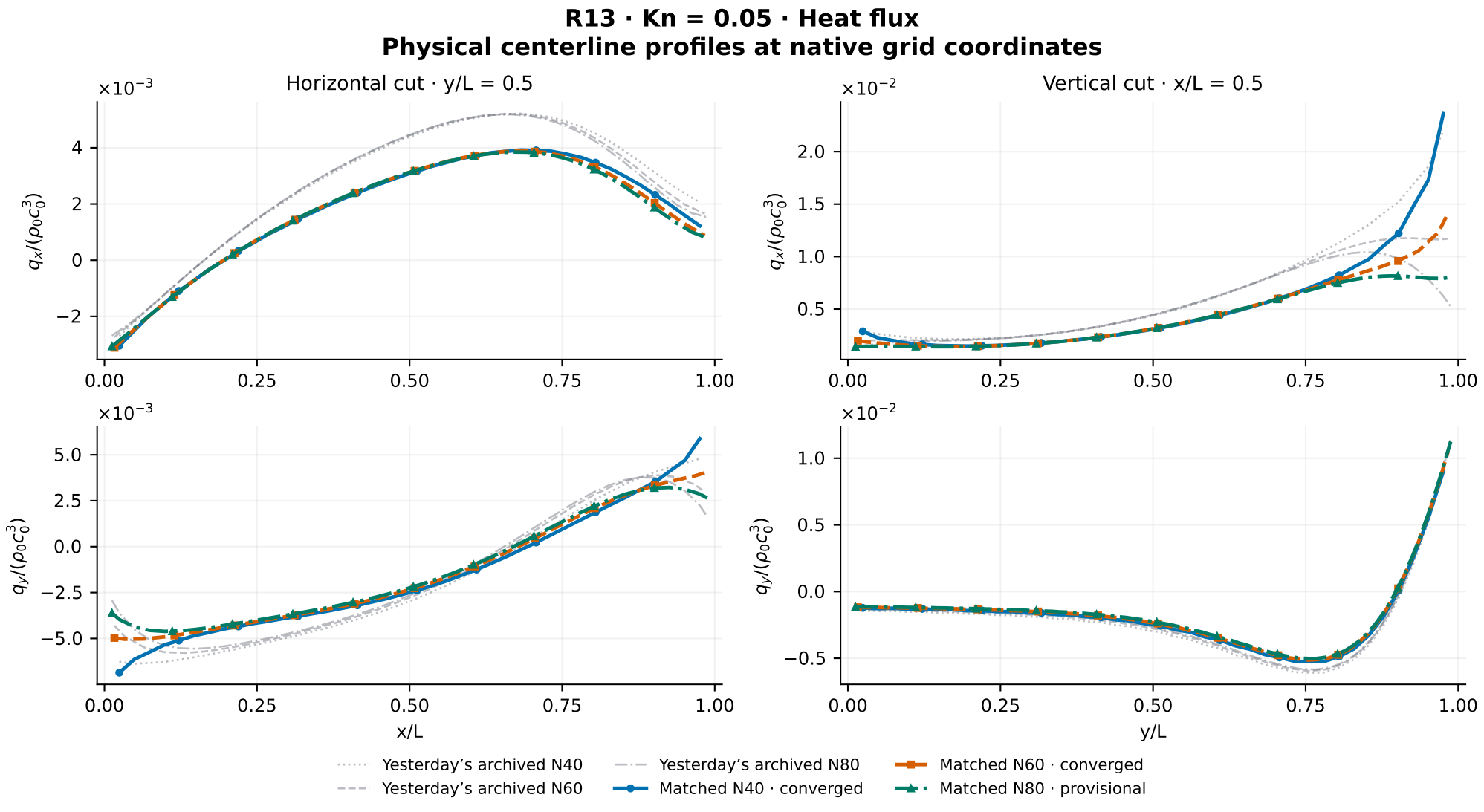
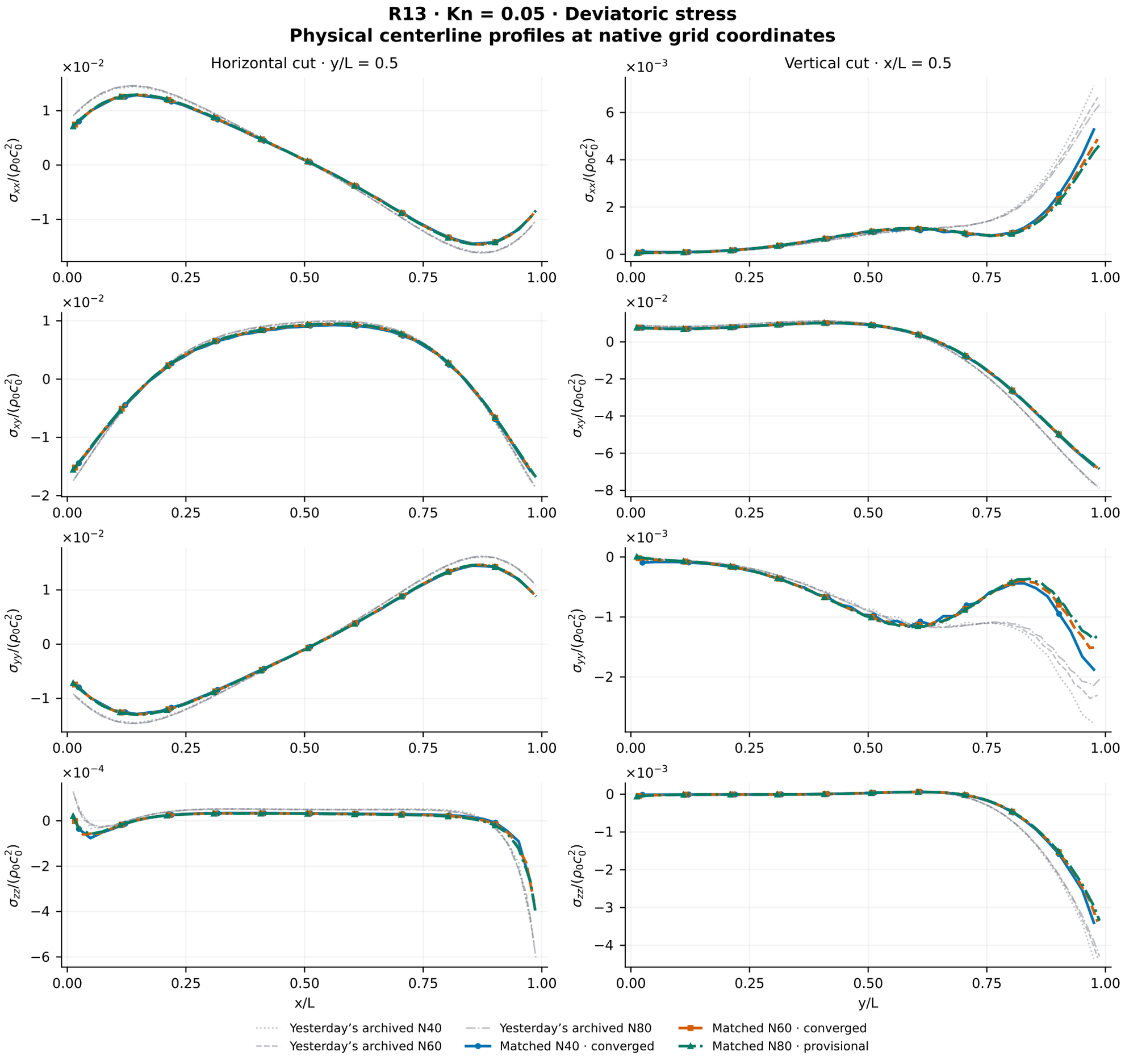
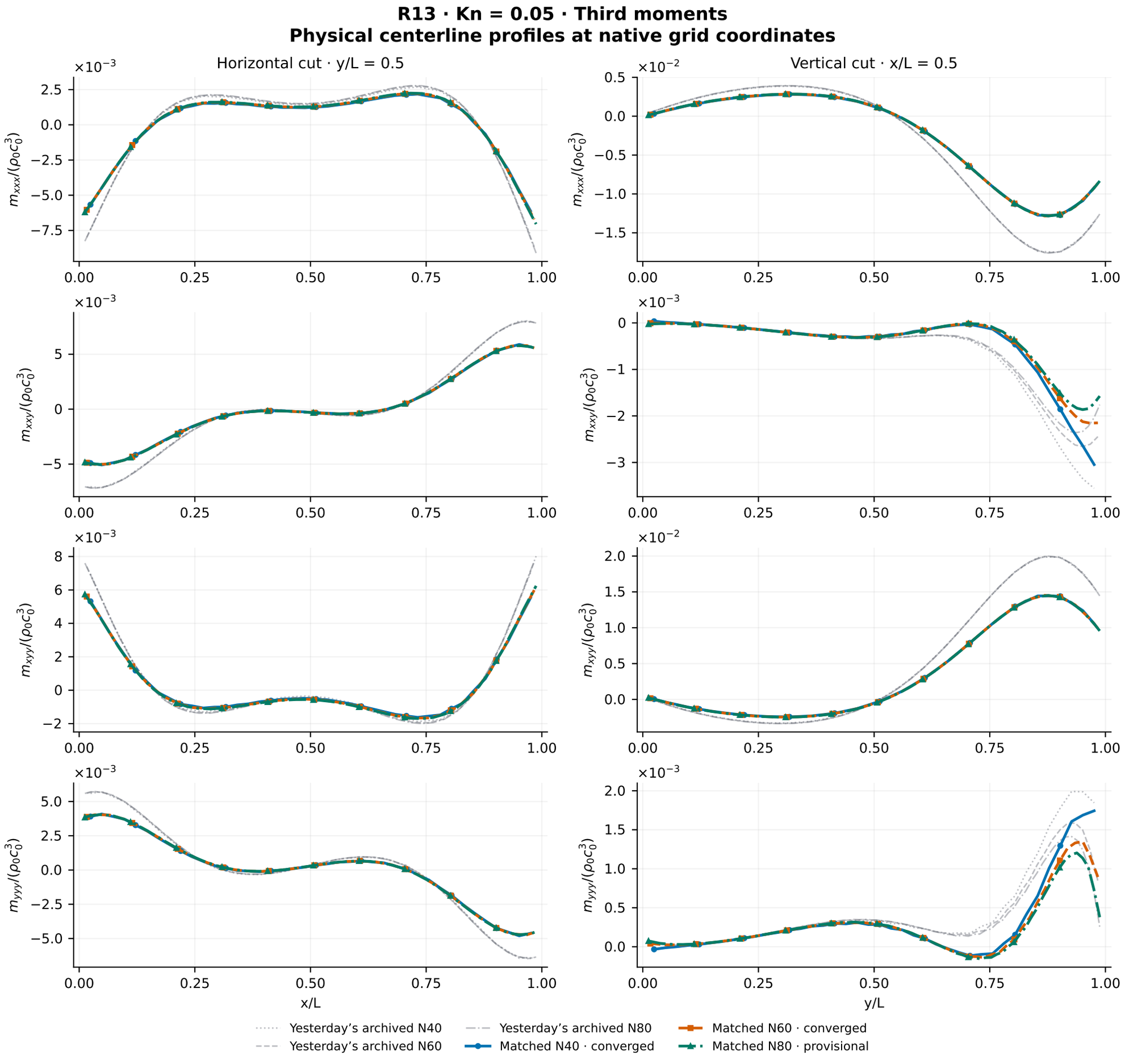
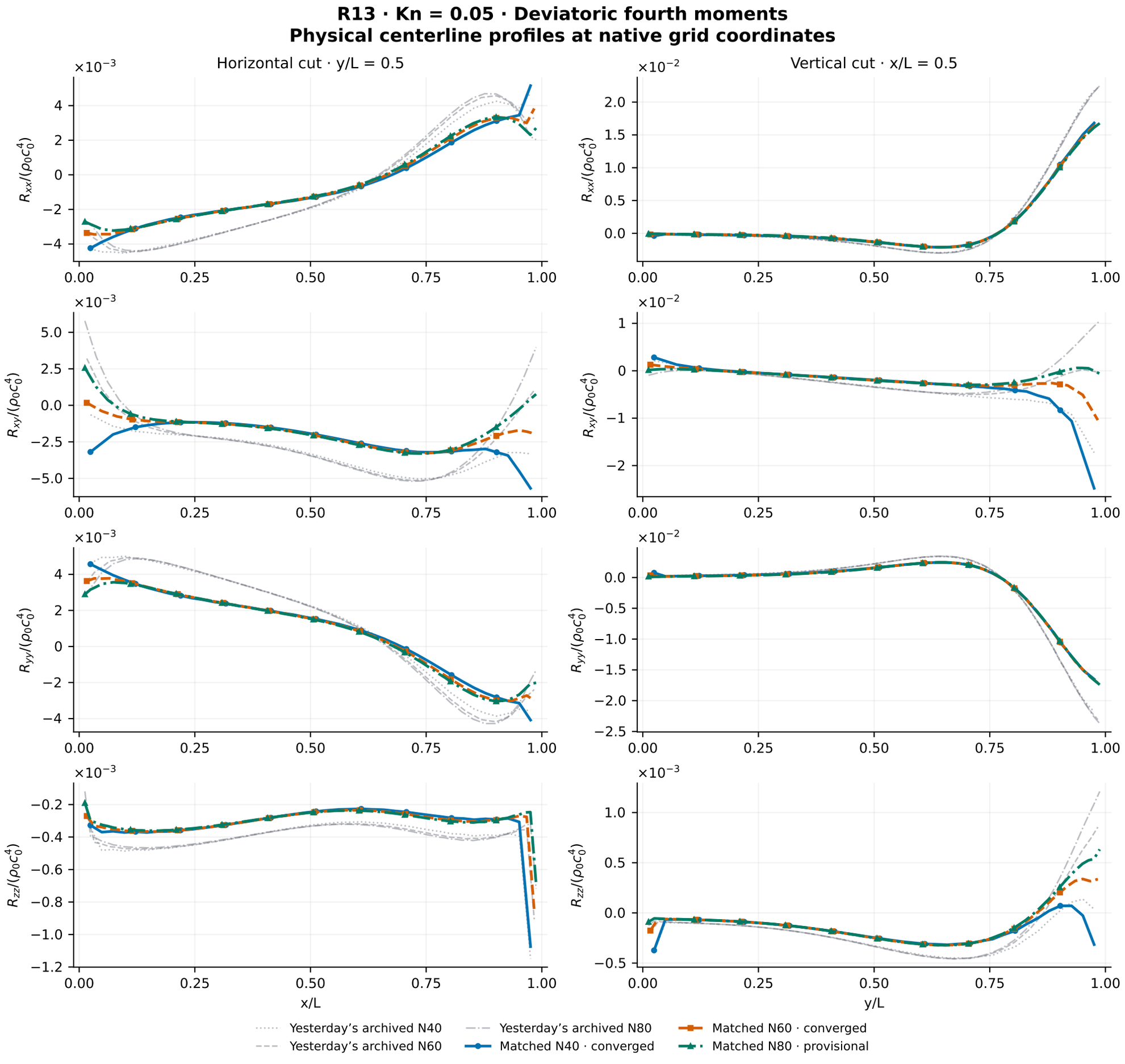

R13 nonlinear convergence
Kn = 0.05
Kn = 0.20
Independent equation residual RMS
Accepted nonlinear step
Accepted nonlinear step
N40 converged
N60 converged
N80 provisional
N40 converged
N60 converged
N80 provisional

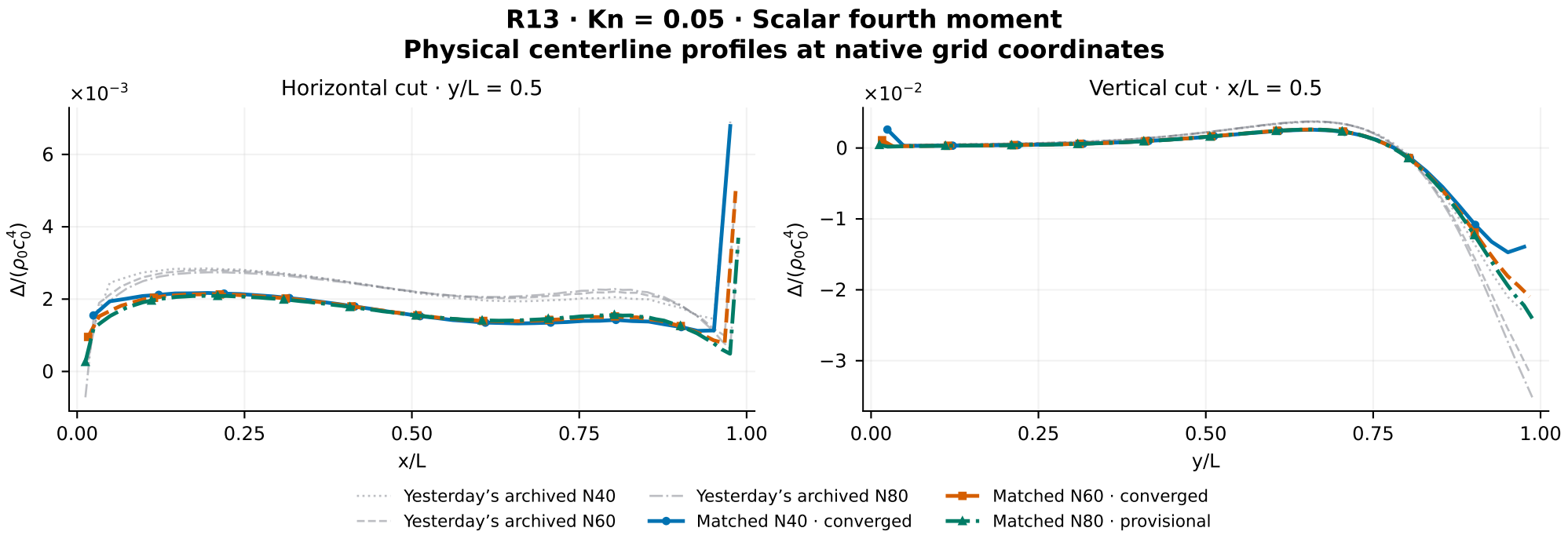

R13 · Kn = 0.05 · Density, velocity and temperature
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
ρ/ρ0
ux/c0
uy/c0
T/T0
×10−2
×10−3
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80

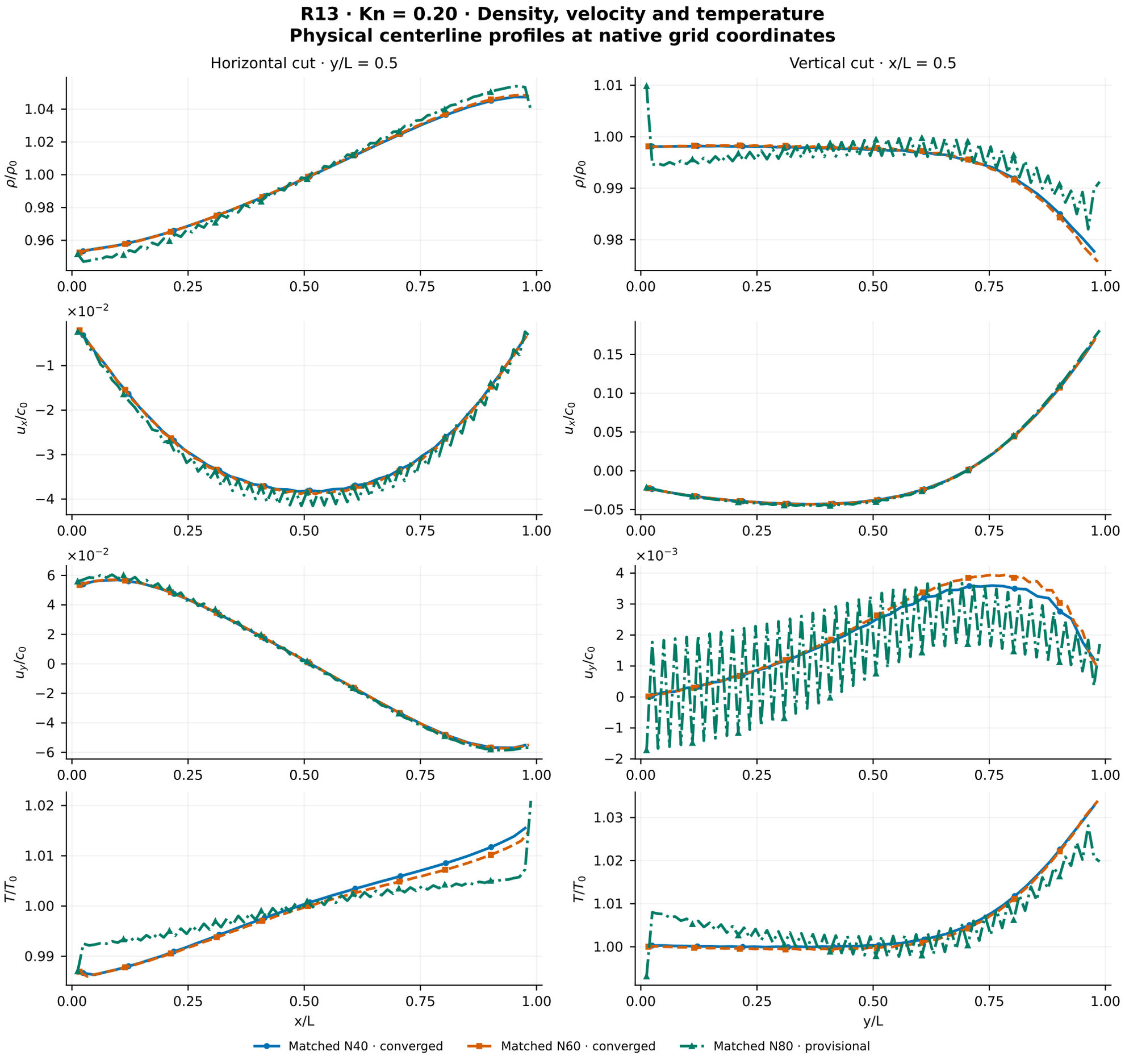

R13 · Kn = 0.05 · Heat flux
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
×10⁻³
×10⁻²
$q_x/(\rho_0 c_0^3)$
$q_y/(\rho_0 c_0^3)$
x/L
y/L
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional

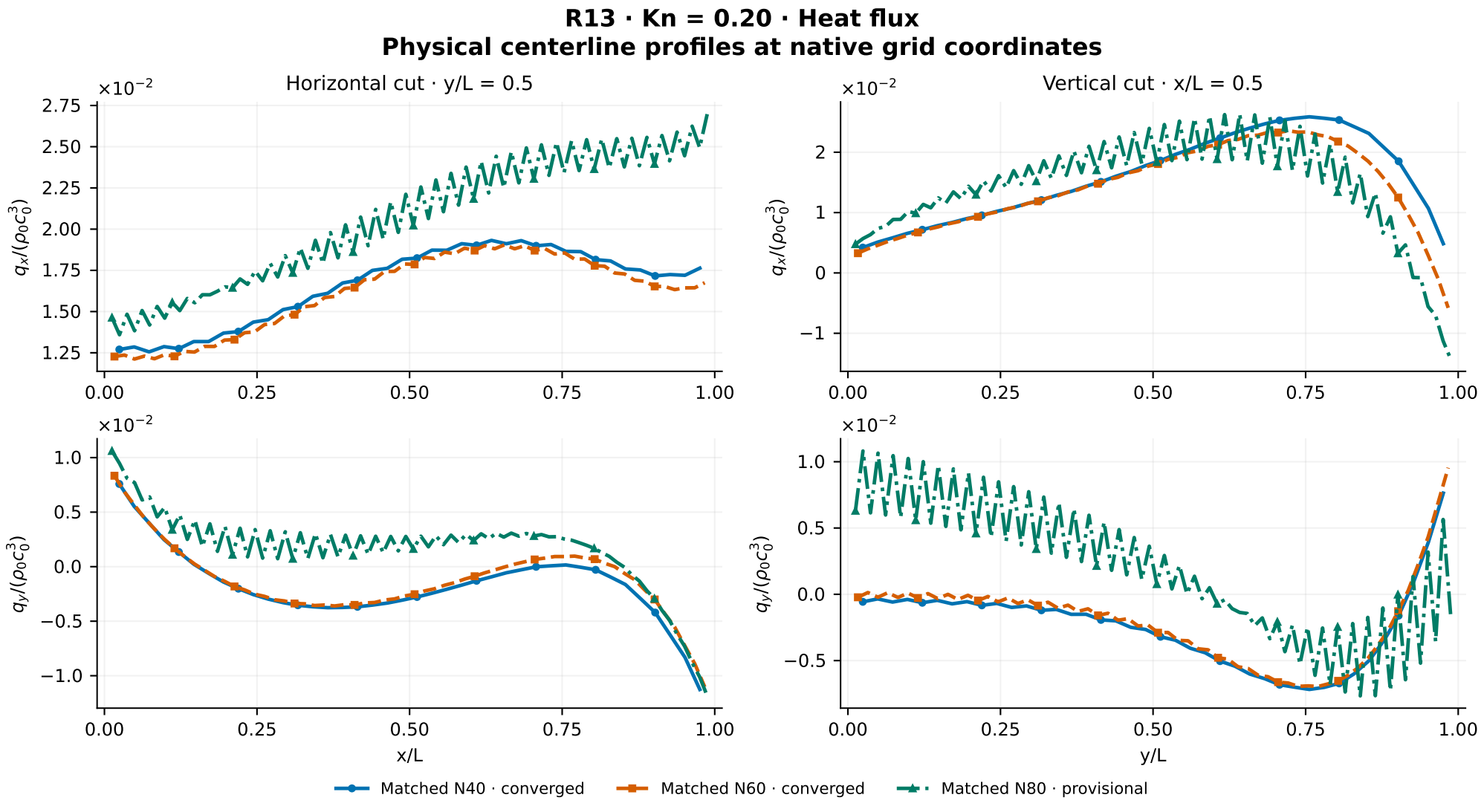

R13 · Kn = 0.05 · Deviatoric stress
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
×10⁻²
×10⁻³
×10⁻⁴
σxx/(ρ0c0²)
σxy/(ρ0c0²)
σyy/(ρ0c0²)
σzz/(ρ0c0²)
x/L
y/L
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional

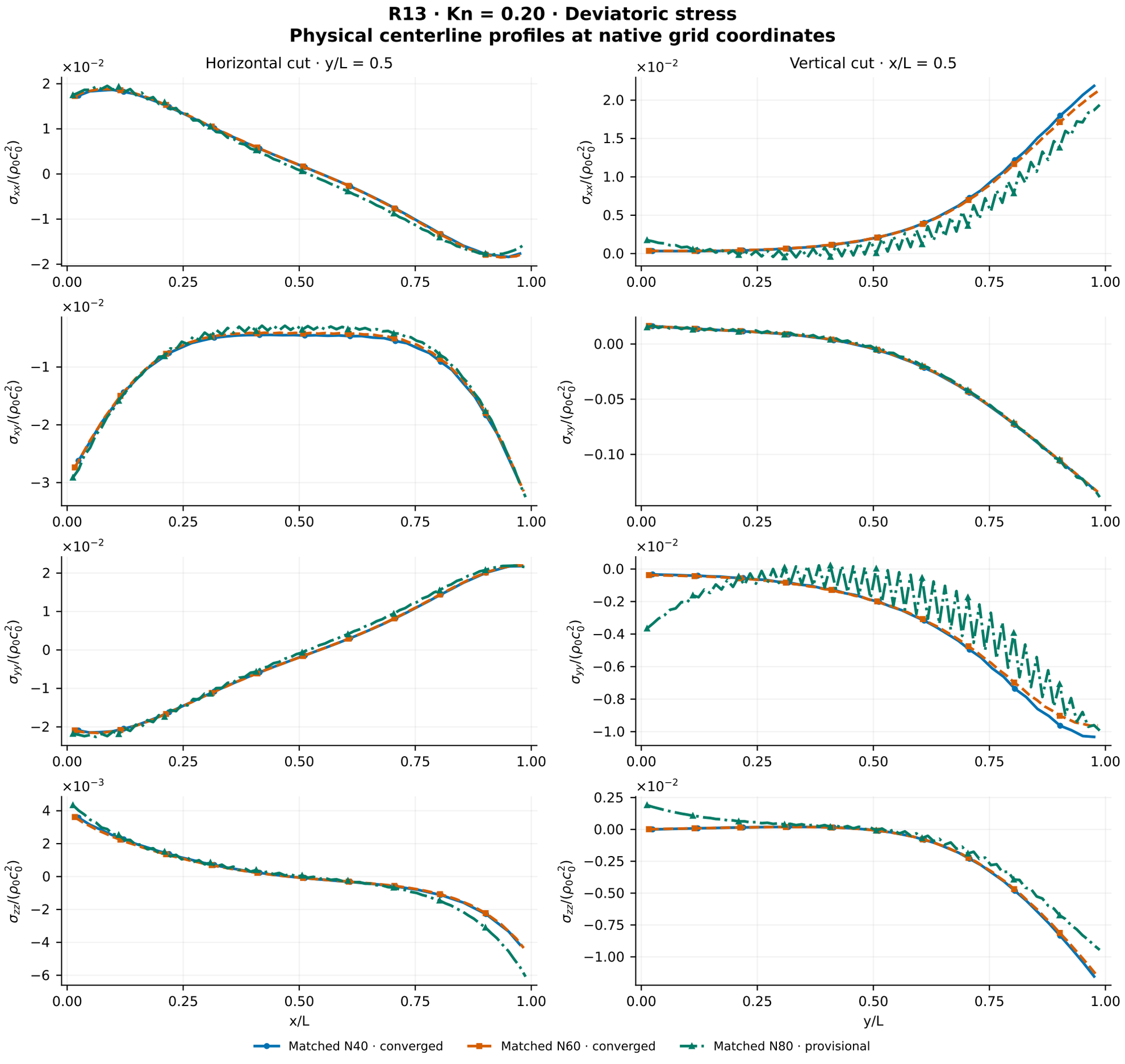

R13 · Kn = 0.05 · Third moments
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
$m_{xxx}/(\rho_0 c_0^3)$
$m_{xxy}/(\rho_0 c_0^3)$
$m_{xyy}/(\rho_0 c_0^3)$
$m_{yyy}/(\rho_0 c_0^3)$
x/L
y/L
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional

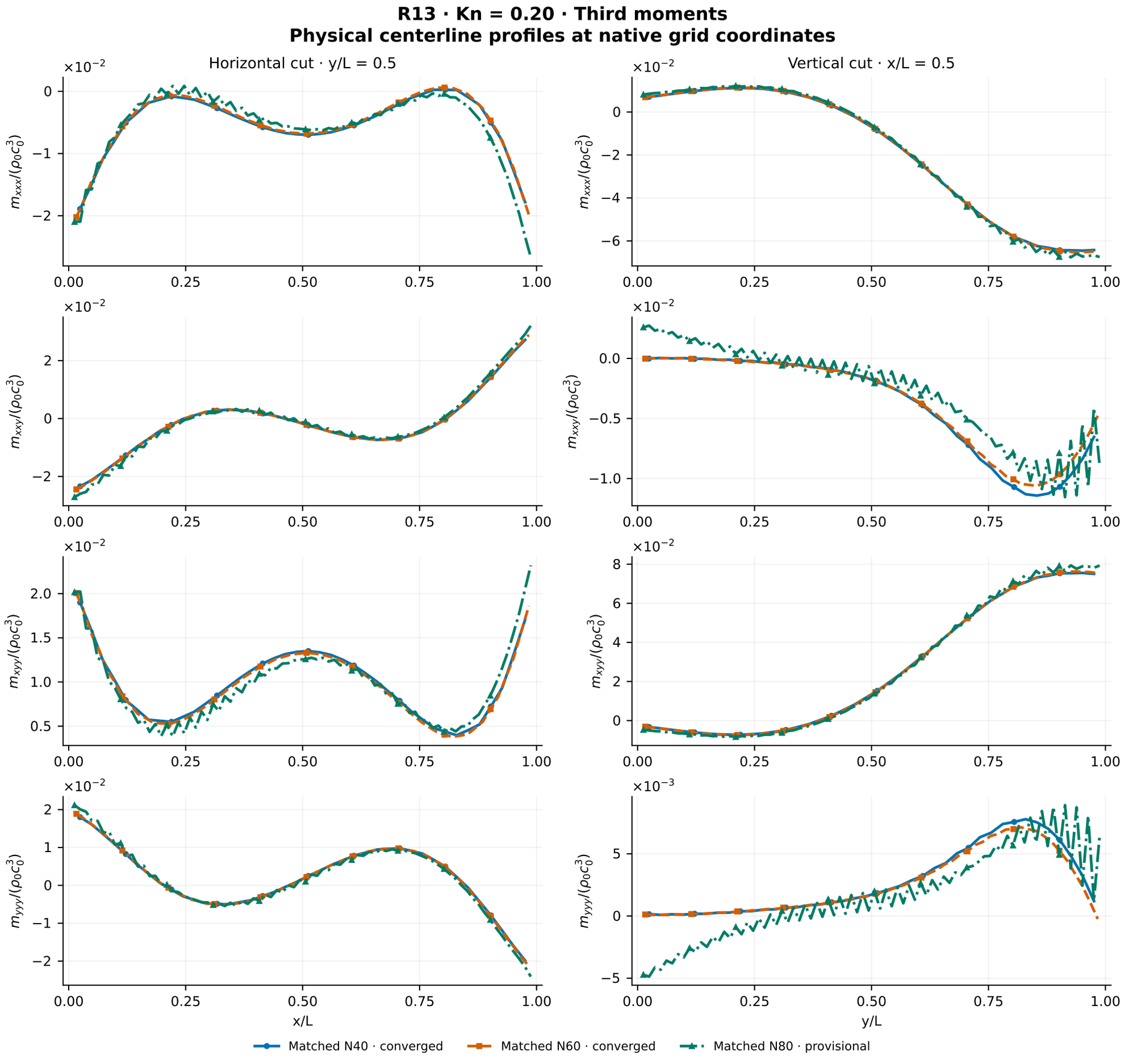

R13 · Kn = 0.05 · Deviatoric fourth moments
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
$R_{xx}/(\rho_0 c_0^4)$
$R_{xy}/(\rho_0 c_0^4)$
$R_{yy}/(\rho_0 c_0^4)$
$R_{zz}/(\rho_0 c_0^4)$
×10⁻³
×10⁻²
x/L
y/L
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional

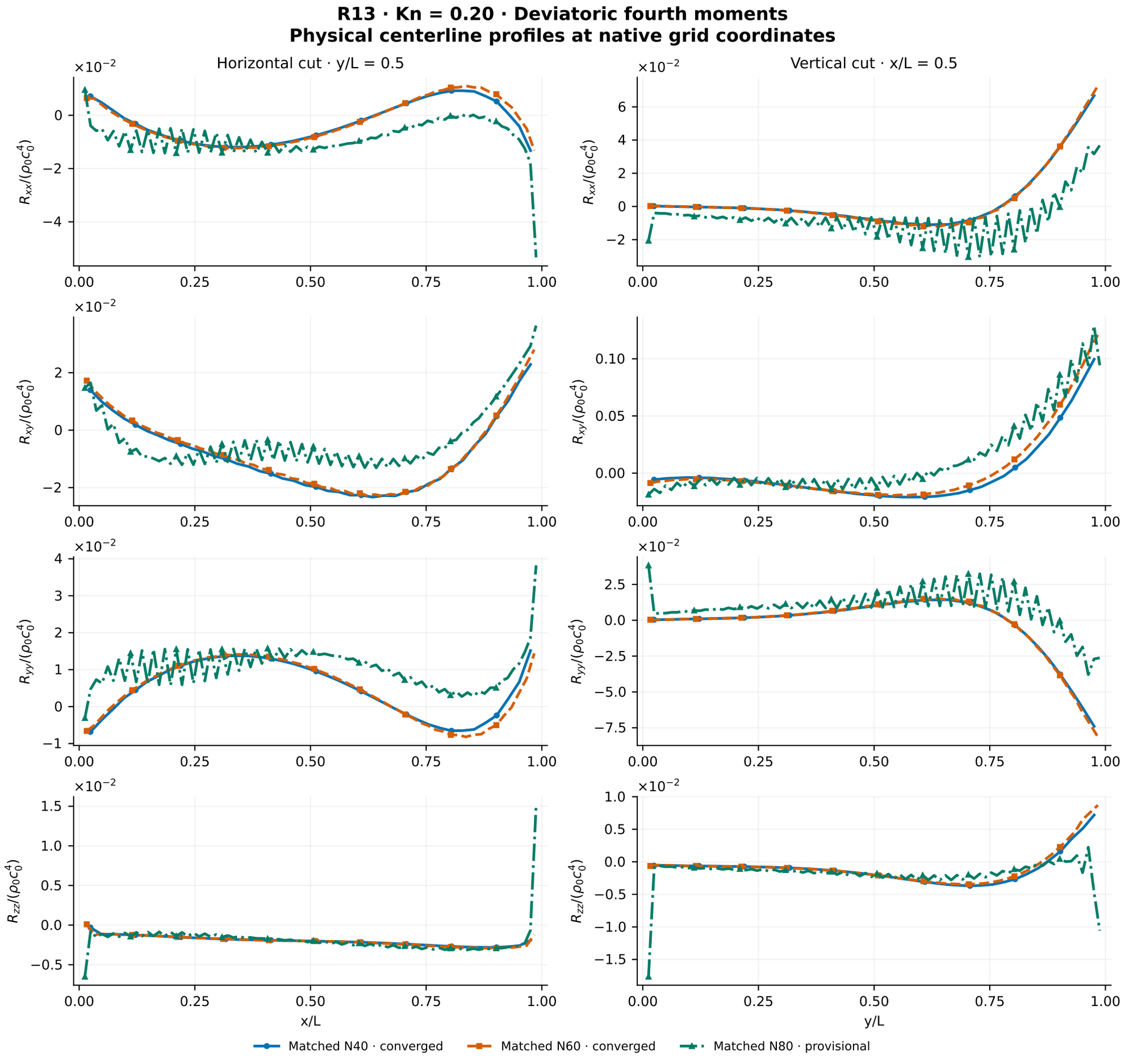

R13 · Kn = 0.05 · Scalar fourth moment
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
×10⁻³
Δ/(ρ₀c₀⁴)
x/L
Vertical cut · x/L = 0.5
×10⁻²
y/L
Yesterday's archived N40
Yesterday's archived N60
Yesterday's archived N80
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional

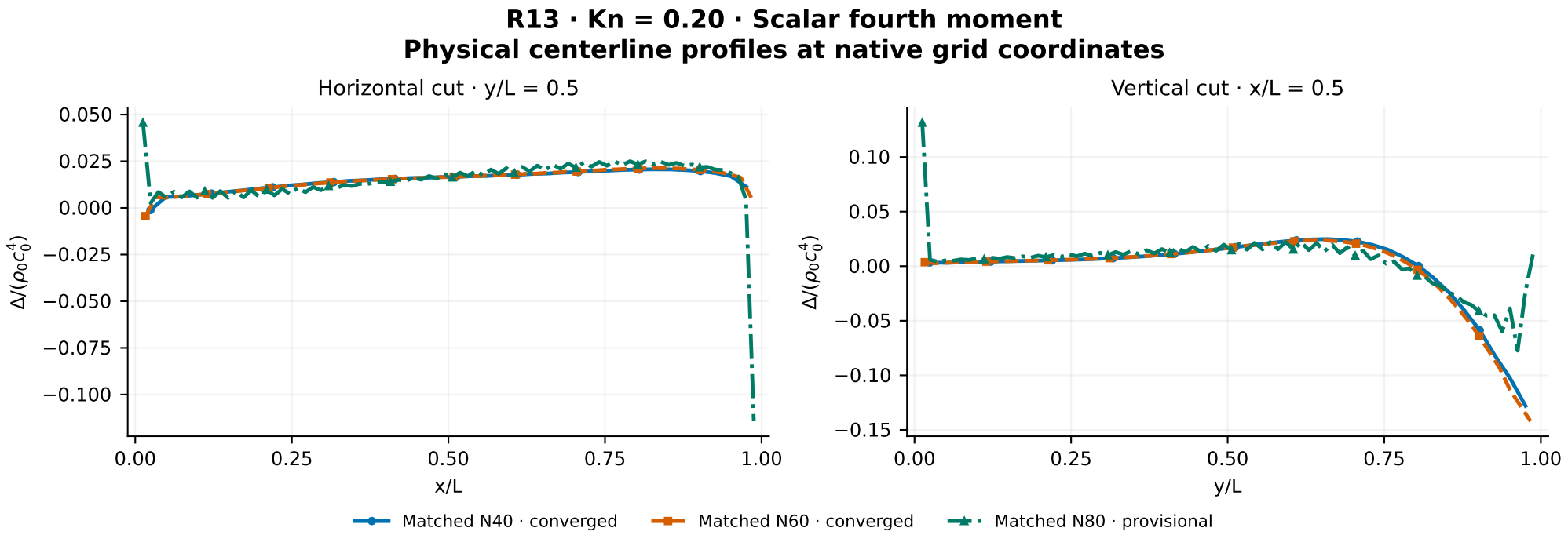

R13 · Kn = 0.20 · Density, velocity and temperature
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
ρ/ρ0
ux/c0
uy/c0
T/T0
×10−2
×10−3
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional


R13 · Kn = 0.20 · Heat flux
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
×10⁻²
$q_x/(\rho_0 c_0^3)$
$q_y/(\rho_0 c_0^3)$
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional


R13 · Kn = 0.20 · Deviatoric stress
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
$\sigma_{xx}/(\rho_0 c_0^2)$
$\sigma_{xy}/(\rho_0 c_0^2)$
$\sigma_{yy}/(\rho_0 c_0^2)$
$\sigma_{zz}/(\rho_0 c_0^2)$
×10⁻²
×10⁻³
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional


R13 · Kn = 0.20 · Third moments
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
$m_{xxx}/(\rho_0 c_0^3)$
$m_{xxy}/(\rho_0 c_0^3)$
$m_{xyy}/(\rho_0 c_0^3)$
$m_{yyy}/(\rho_0 c_0^3)$
×10⁻²
×10⁻³
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional


R13 · Kn = 0.20 · Deviatoric fourth moments
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
$R_{xx}/(\rho_0 c_0^4)$
$R_{xy}/(\rho_0 c_0^4)$
$R_{yy}/(\rho_0 c_0^4)$
$R_{zz}/(\rho_0 c_0^4)$
×10⁻²
x/L
y/L
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional


R13 · Kn = 0.20 · Scalar fourth moment
Physical centerline profiles at native grid coordinates
Horizontal cut · y/L = 0.5
Vertical cut · x/L = 0.5
Δ/(ρ0c0^4)
x/L
y/L
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.00
0.25
0.50
0.75
1.00
Matched N40 · converged
Matched N60 · converged
Matched N80 · provisional